

HYUNG GYU MIN

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Fields of Interest

Single-cell genomics Functional genomics Sequence data analysis Dimension reduction Quantum computing

Education

Oct 2023 - Present	UNC at Chapel Hill , Chapel Hill, North Carolina PhD, Biostatistics
Sep 2020 - Oct 2023	Yonsei University , Seoul, Republic of Korea Master of Arts, Statistics GPA: 4.27/4.30 (4.00/4.00)
Mar 2014 - Aug 2020	Yonsei University , Seoul, Republic of Korea Bachelor of Arts, Applied Statistics Stat & Math GPA: 4.11/4.30 (3.91/4.00) Overall GPA: 3.70/4.30 (3.56/4.00)

Publications

Functional Genomics Published	Massively parallel reporter assays investigate shared genetic variants across eight psychiatric disorders (2025) <i>Cell</i> , contributing author
Single-Cell Genomics Under review	Perturbation of genes linked to common schizophrenia risk variants identifies cilia-associated gene programs <i>Cell</i> , second author
Clinical Statistics Published	Effect of orally administered heat-treated <i>Lactobacillus plantarum</i> LM1004 on the innate immune system <i>Journal of Functional Foods</i> , contributing author
Microbiome Analysis Published	Altered gut microbiota in individuals with episodic and chronic migraine <i>Scientific Reports</i> , contributing author

Teaching Experience

Yonsei University	Teaching Assistant, Quantum Finance Computing (Sep 2022 – Dec 2022) Teaching Assistant, Multivariate Analysis (Sep 2022 – Dec 2022) Teaching Assistant, Theory of Dimension Reduction (Mar 2022 – Jun 2022) Teaching Assistant, Qiskit for Data Science (Mar 2022 – Jun 2022) Teaching Assistant, Big Data and Quantum Computing (Sep 2021 – Dec 2021) Teaching Assistant, Multivariate Analysis (Sep 2021 – Dec 2021) Teaching Assistant, Statistical Data Analysis (Mar 2021 – Jun 2021) Instructor: Prof. Hakbae Lee
Yonsei University	Teaching Assistant, Introduction to Statistics (Sep 2020 – Dec 2020) Instructor: Prof. Hogyu Lee
Korea Quantum Computing	Lecturer, Introduction to Quantum Data Science (Jan 2023 – Feb 2023) Director: Dr. Kion Kim

Research & Leadership Experience

UNC Chapel Hill **Won Lab, Department of Genetics** (Oct 2023 – Present)
Ph.D. Researcher Research in single-cell and functional genomics, including in vivo MPRA and scMPRA
Development of statistical and computational methods for high-dimensional genomic data
Advisor: Prof. Hyejung Won, Web : <https://www.wonlab.org/>

Korea Quantum Computing **R&D Department** (Jun 2022 – May 2023)
Researcher (Assistant Manager) IBM Hub activities for quantum computing outreach and research in quantum data science
Conducted projects in quantum data science and delivered introductory lectures on quantum computing
Supervisor: Dr. Kion Kim

Yonsei University **Quantum DnA Laboratory** (Mar 2022 – Feb 2023)
Researcher Research on quantum convolutional neural networks and quantum error mitigation
Advisor: Prof. Daniel Kyungdeock Park

Yonsei University **Institute of Data Science** (Mar 2021 – Aug 2023)
Chief Consulting Assistant Led statistical consulting projects across disciplines (biology, social science, engineering, etc.)
Consulted 20+ cases in 2021; led junior consultants and supervised projects in 2022
Director: Prof. SeungHo Kang

Yonsei University **Data Science Lab (DSL)** (Jan 2019 – Dec 2020)
Director of Academic Affairs Undergraduate academic society focused on data science and statistical theory
Founding member; designed study programs and coordinated projects
Advisor: Prof. TaeYoung Park

Programming Skills

Programming: Python (Mudata, Dask, FAISS, scikit-learn, PyTorch/TensorFlow),
R (Seurat, Rcpp, Shiny, tidyverse), SAS, Linux, Slurm
Tools & Platforms: Docker, Git, SQL, Notion, Slack

Awards & Honors

May 2019 Excellence Prize, 2019 Embrain Panel Big Data Analysis Competition
Fall 2019 High Honors, Yonsei University